INCH-POUND

MIL-DTL-25666/2 (USAF)
15 September 1995
SUPERSEDING
MIL-C-25781C (USAF)
27 March 1981

DETAIL SPECIFICATION

CONVERTER, LIQUID OXYGEN GCU-3/A

This specification is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of converter. It provides storage and pressurization of 10 liters of liquid oxygen to 70 pounds per square inch gauge (psig), nominal pressure and does not have integral warming coils. The converter is designated GCU-3/A.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are needed to meet the requirements specified in sections 3, 4, and 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3, 4, and 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index for Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-V-9050	Valves, Oxygen, Pressure Relief, Aircraft
MIL-C-25516	Connectors, Electrical, Miniature, Coaxial, Environment Resistant Type, General Specification For
MIL-DTL-25666	Converters, Liquid Oxygen, Capacitance Type Gauging, General Specification For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: OC-ALC/TICLA, 3001 Staff Drive, Tinker AFB, OK 73145-3037 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1660

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STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-129 Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, and standards are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Ave, Philadelphia, PA 19111-5094)

2.3 Order of Precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this documents takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The converters furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time set for opening of bids (see 4.2 and 6.3).

3.2 Design and construction. The converter shall conform to figures 1, 2 and 3. The converter shall have an operating pressure of 70 pounds per square inch gage (psig) and a minimum capacity of 10 liters of liquid oxygen measured at 1 atmosphere.

3.2.1 Strength. The filled converter shall be designed to withstand the dynamic loads that are imposed by the following accelerations:

Vertical (down) 14G

Vertical (up) 6G

Lateral 6G

3.2.2 Marking of ports. The ports shall be marked as indicated in figure 1.

3.2.3 Pressure relief valves. Tile pressure relief valves shall conform to type II of MIL-V-9050 and shall be of such size and shape to fit the over-all dimensions specified for the converter.

3.2.4 Weight. The weight of the converter shall not exceed 25 pounds.

3.3 Performance.

3.3.1 Final assembly pressure. The final assembly pressure of the converter shall be 95 psig.

3.3.2 Filling time. The time required to fill the converter shall not exceed 5 minutes.

3.3.3 Evaporation loss.

3.3.3.1 Vent condition. The evaporation loss of the essentially full converter in the vent condition shall not exceed 3 pounds of oxygen when tested as specified in MIL-DTL-25666.

3.3.3.2 Buildup condition. The evaporation loss of the essentially full converter in the buildup condition shall not exceed 3 pounds of oxygen when tested as specified in MIL-DTL-25666.

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3.3.4 Buildup time. The time required for the converter to reach its operating pressure immediately after it has been filled shall not exceed 10 minutes.

3.3.5 Capacitance. The capacitance of the sensing element, after it is installed in the empty converter, shall be 123.5 ±1.0 picofarad (PF).

3.3.6 Inner vessel pressure.

3.3.6.1 Inner vessel. The inner vessel shall meet the performance requirements for the inner vessel pressure test specified in MIL-DTL-25666 when subjected to a pressure of 200 psig.

3.3.6.2 Permanent volumetric expansion. The converter shall meet the performance requirements for permanent volumetric expansion as specified in MIL-DTL-25666 when subjected to a pneumatic or hydrostatic pressure of 200 psig .

3 3.7 Delivery rate.

3.3.7.1 Constant flow. The converter shall deliver a flow of gaseous oxygen at 72 ±10 liters per minute for not less than 1.5 hours with the operating pressure remaining between 70 and 80 psig after the first 15 minutes of flow. Pressures in the supply line shall remain between 55 psig and 105 psig during the entire period of flow. The flow at the specified rate shall be initiated within 10 minutes after the converter has reached the operating pressure. The full test procedure is specified in MIL-DTL-25666.

3.3.7.2 Cyclic flow. The converter shall deliver oxygen as specified (see 3.4.7.1) for not less than 1.5 hours when tested as specified in MIL-DTL-25666.

3.3.8 Calibration. The capacitance sensing element shall have characteristics that provide for a linear relationship between electrical capacitance and volume of liquid oxygen sensed. The relationship shall be in accordance with figure 4, using the equations that are specified (see 3.3.8.1) to determine tolerances. An example of the use of these equations is as follows: If the actual quantity of liquid oxygen at the 6 liter test point is 6.1 liters, the measured capacitance must be between C minimum (min) and C maximum (max).

C min = 5.7 x 6.1 + 122.3 = 157.1 PF
C max = 5.9 x 6.1 + 124.7 = 160.7 PF

3.3.8.1 Equations. The following equations shall be used to calculate the minimum and the maximum capacitances.

Values less than 2 liters	Values greater than 2 liters
C min = 5.6 L + 122.5	C min = 5.7 L + 122.3
C max = 6.0 L + 124.5	C max = 5.9 L + 124.7

C = Capacitance in PF
L = Exact quantity in liters of liquid at test point

3.3.9 Operating temperature. The operating pressure shall remain between 70 and 85 psig after the first 15 minutes of flow at the low temperature and shall remain between 65 and 80 psig after the first 15 minutes of flow at the high temperature. An intermittent increase or decrease of pressure is allowable provided these variances do not exceed a 2 minute interval for any one pressure cycle.

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3.3.10 Vibration.

3.3.10.1 Vibration.. The converter shall not exhibit physical evidence of damage after completion of the vibration test specified in MIL-DTL-25666

3.3.10.2 Vibration - resonance. The converter shall not exhibit physical evidence of damage after completion of the resonance portion of the vibration test specified in MIL-DTL-25666.

3.3.10.3 Vibration - cycling. The converter shall not exhibit physical evidence of damage after completion of the cycling portion of the vibration test specified (see 4.4.1).

3.4 Check valve. The check valve used in the system shall be designed to open at 5 ±2 psig.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2)

b. Conformance inspection (see 4.3).

4.2 Qualification inspection. The qualification inspection shall consist of all of the methods of inspection that are d specified in MIL-DTL-25666.

4.3 Conformance inspection. The conformance inspection shall be in accordance with MIL-DTL-25666.

4.4 Vibration.

4.4.1 Vibration - cycling. The converter shall be vibrated under the cycling conditions specified herein for the applicable periods listed in table I. The frequency shall be cycled between 5 to 500 cps and return in a 15 minute cycle at an applied double amplitude of 0.036 inch or an applied acceleration of ±10G which ever is the lower G value. The rate of change of frequency shall be logarithmic (see 3.3.10.3).

5. PACKAGING

5.1 Packaging. Packaging requirements shall be as specified in the contract or order (see 6.2).

5.2 Marking. Unless otherwise specified (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The 10 liter, 70 psig converter covered by this specification is intended to be used on the F-106A/B aircraft.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

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6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL-25666 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from, OC-ALC/TICLA, 3001 Staff Drive, Tinker AFB, OK 73145-5990.

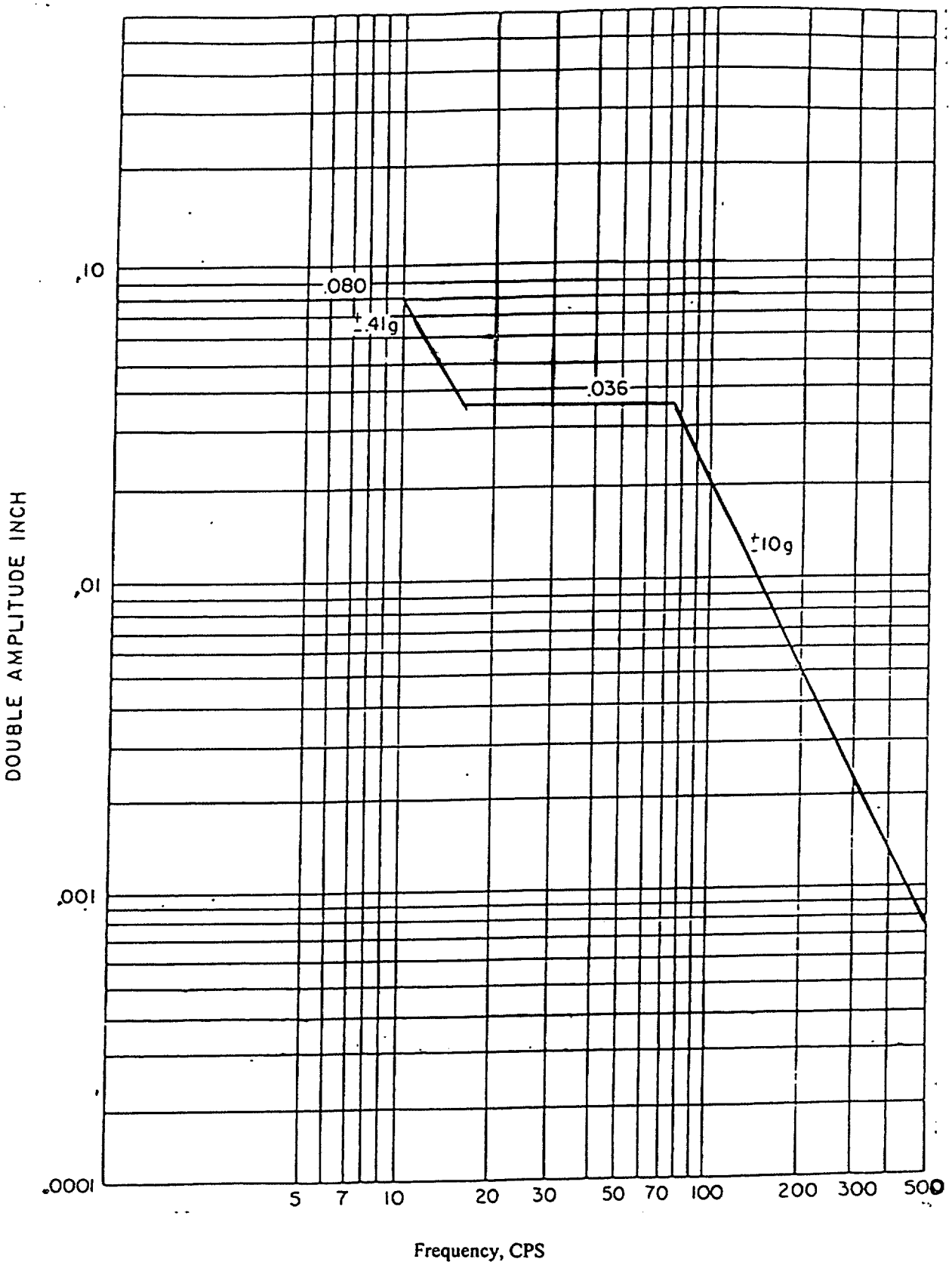
6.4 Subject term (key word) listing.

Breathing oxygen
Capacitance
Container, inner
Container, outer
Delivery rate
Dewar
Element, sensing
Evaporation loss
Gunfire
Life support
LOX
Valve, fill-buildup-vent
Valve, pressure closing
Valve, pressure relief
Vibration

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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TABLE I. Range curve for converter vibration



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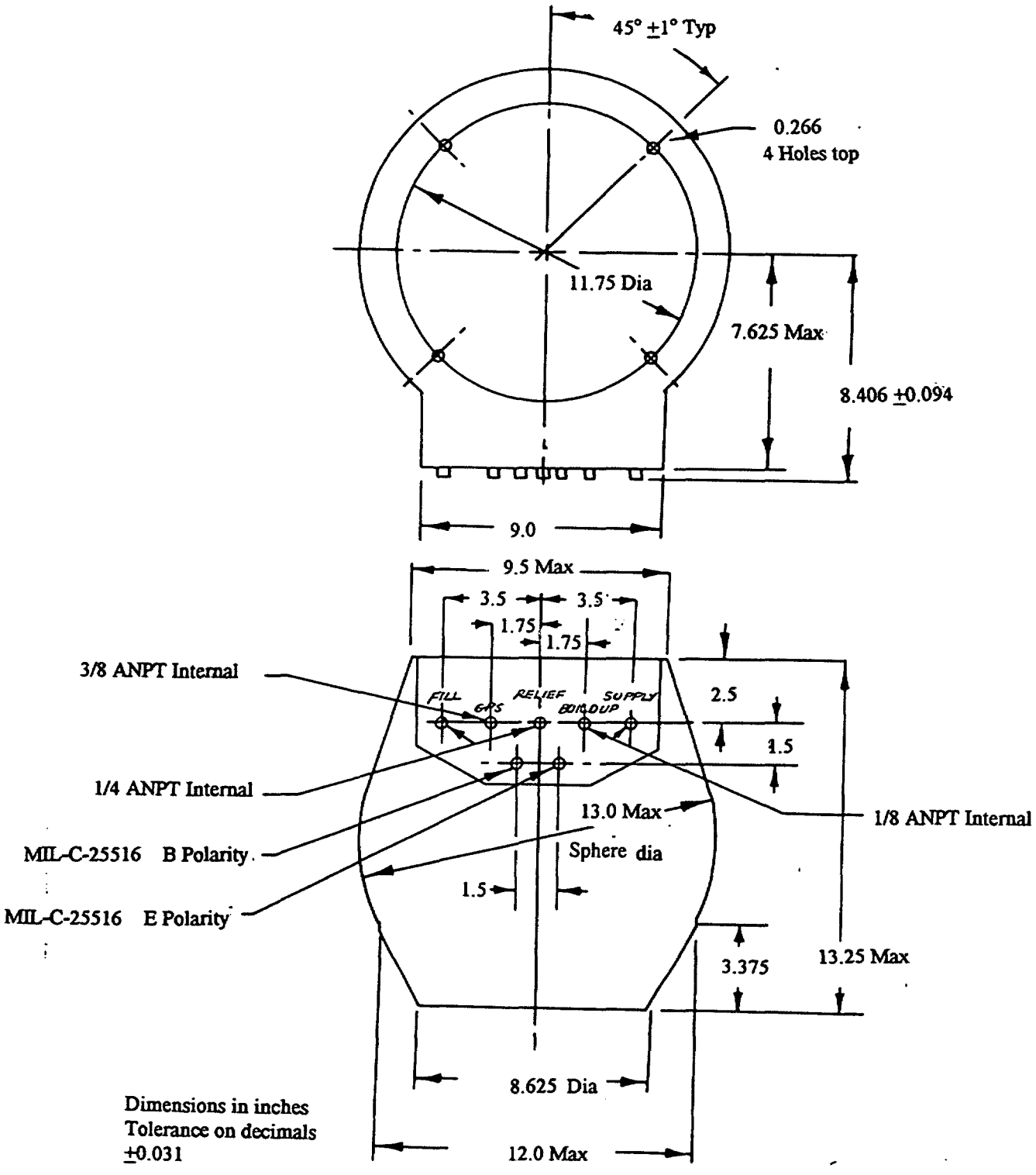
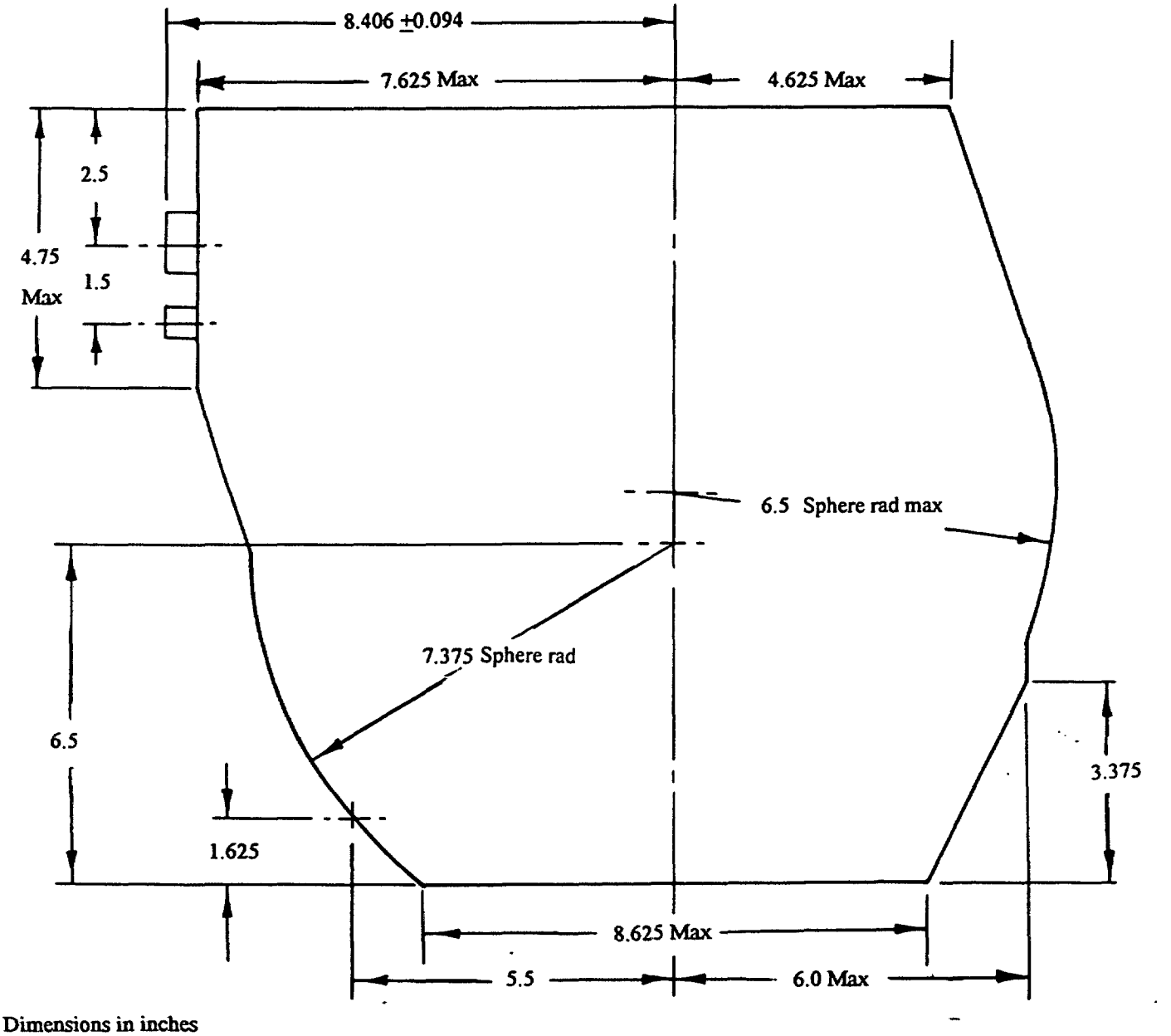


FIGURE 1. Top and front view

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Dimensions in inches
Tolerance on decimals ±0.031

FIGURE 2. Side view

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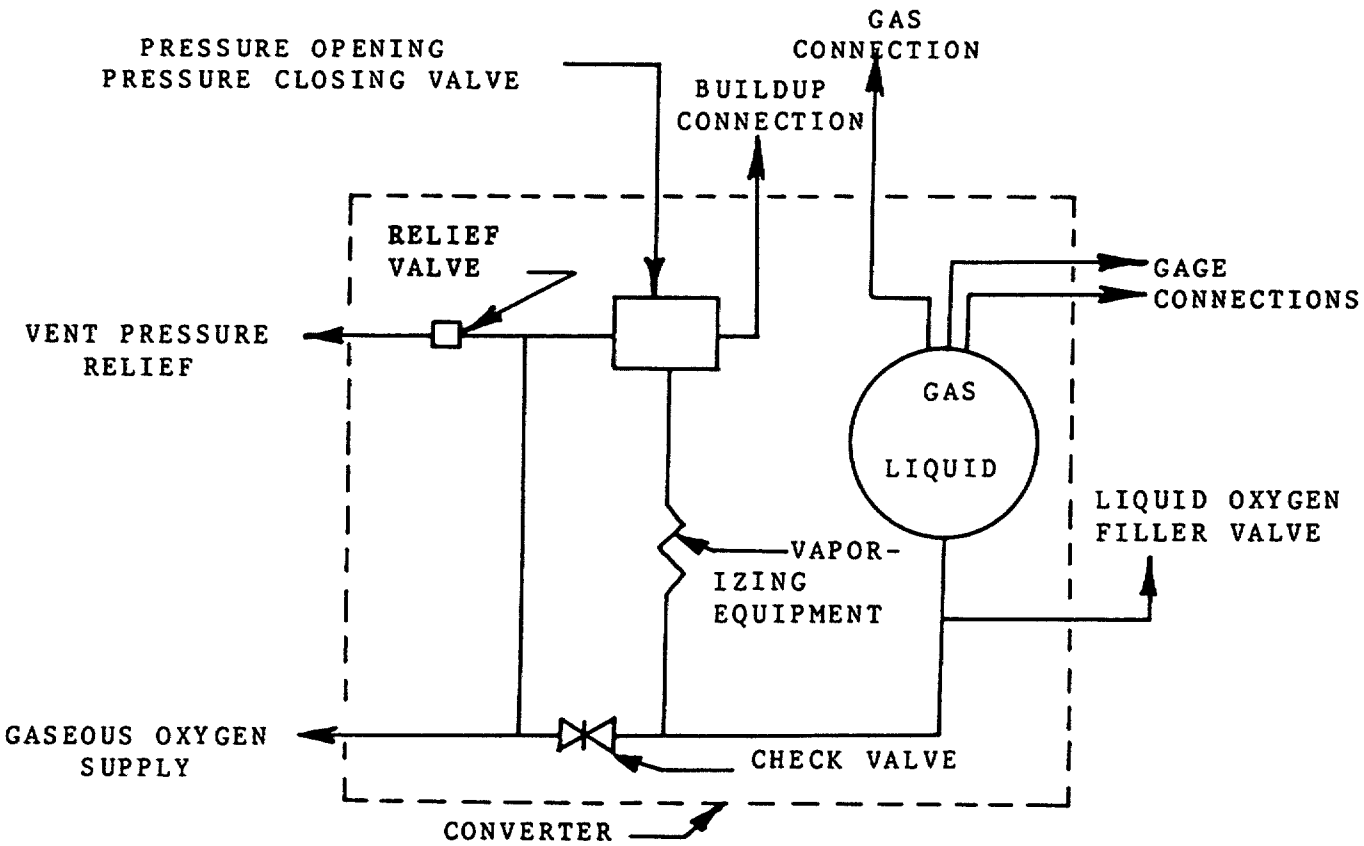
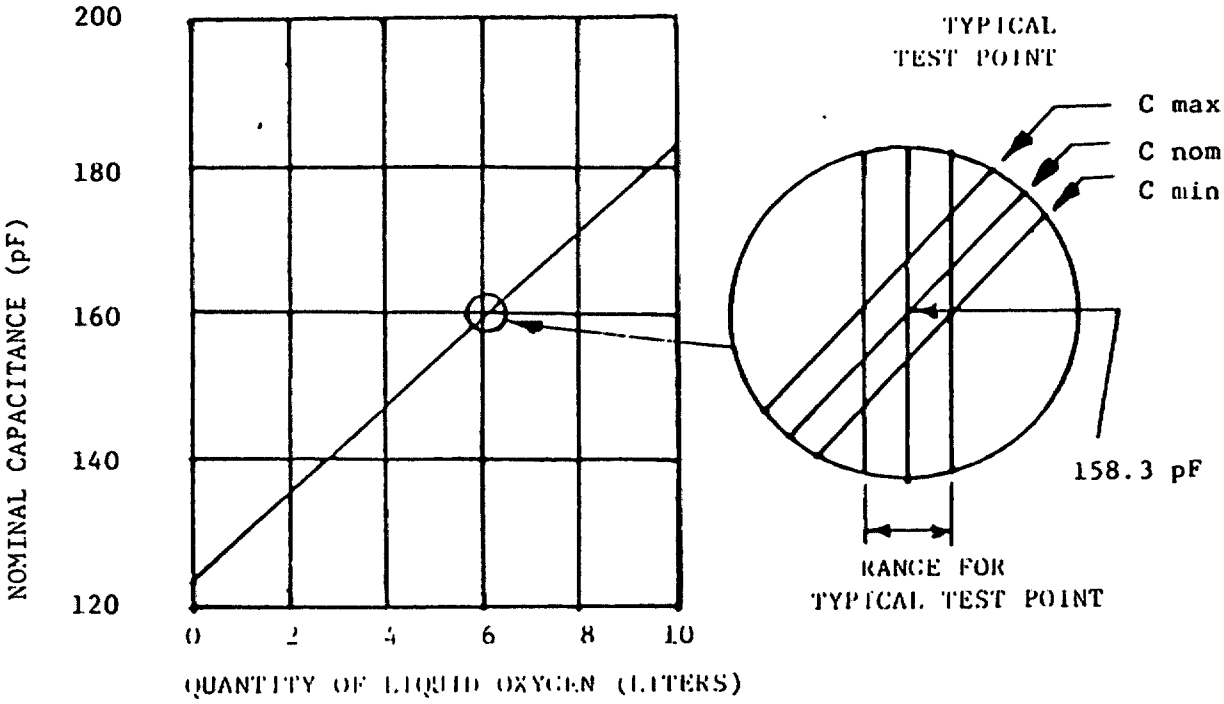


FIGURE 3. Schematic diagram

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Liquid oxygen (liters)	Nominal capacitance (pf)
0	123.5
2	135.1
4	146.7
6	158.3
8	169.9
10	181.5

FIGURE 4. Relationship between capacitance and volume of oxygen sensed

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	MIL-DTL-25666/2	950915

3. DOCUMENT TITLE: CONVERTER, LIQUID OXYGEN GCU-3/A

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

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